

A Appendix

A.1 Detailed model architecture

Figures F1, and F2, F3 show the detailed architecture of the multimodal encoder in Pendulum, Hyperhot and VizDoom, respectively.

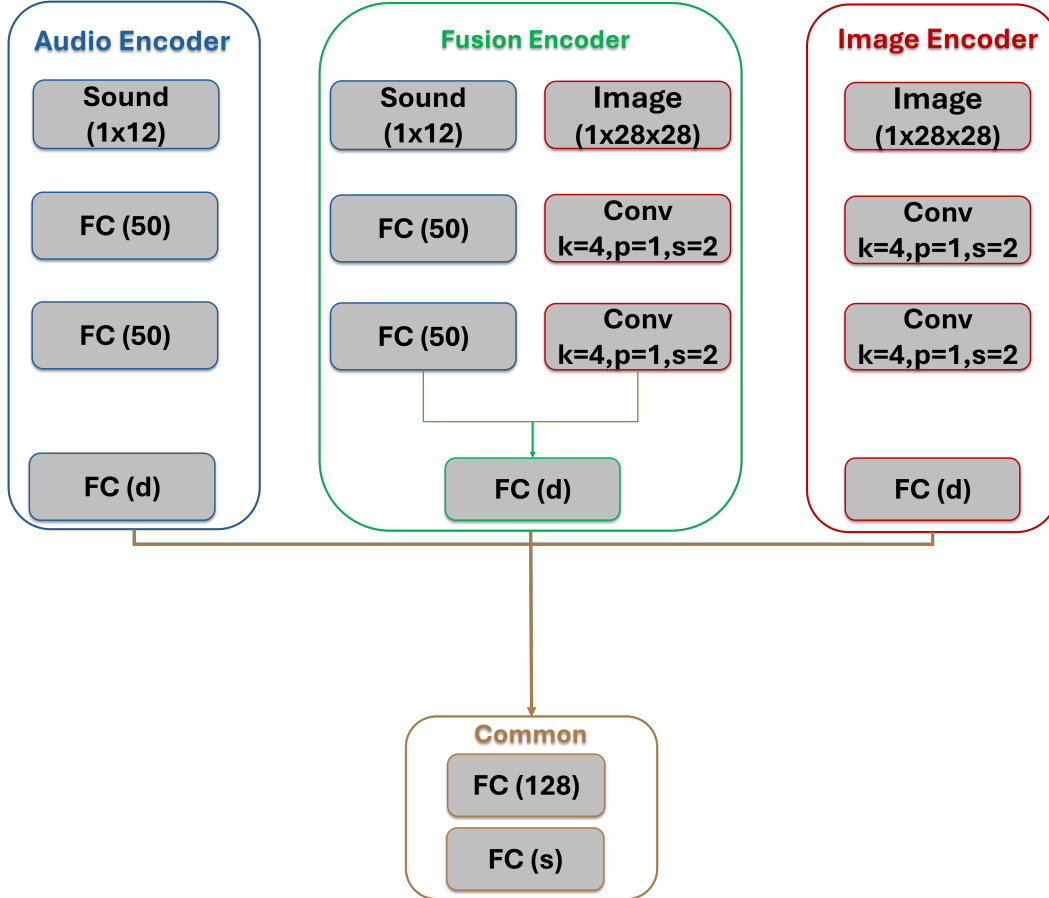


Figure F1: Multimodal encoder architecture used in PENDULUM

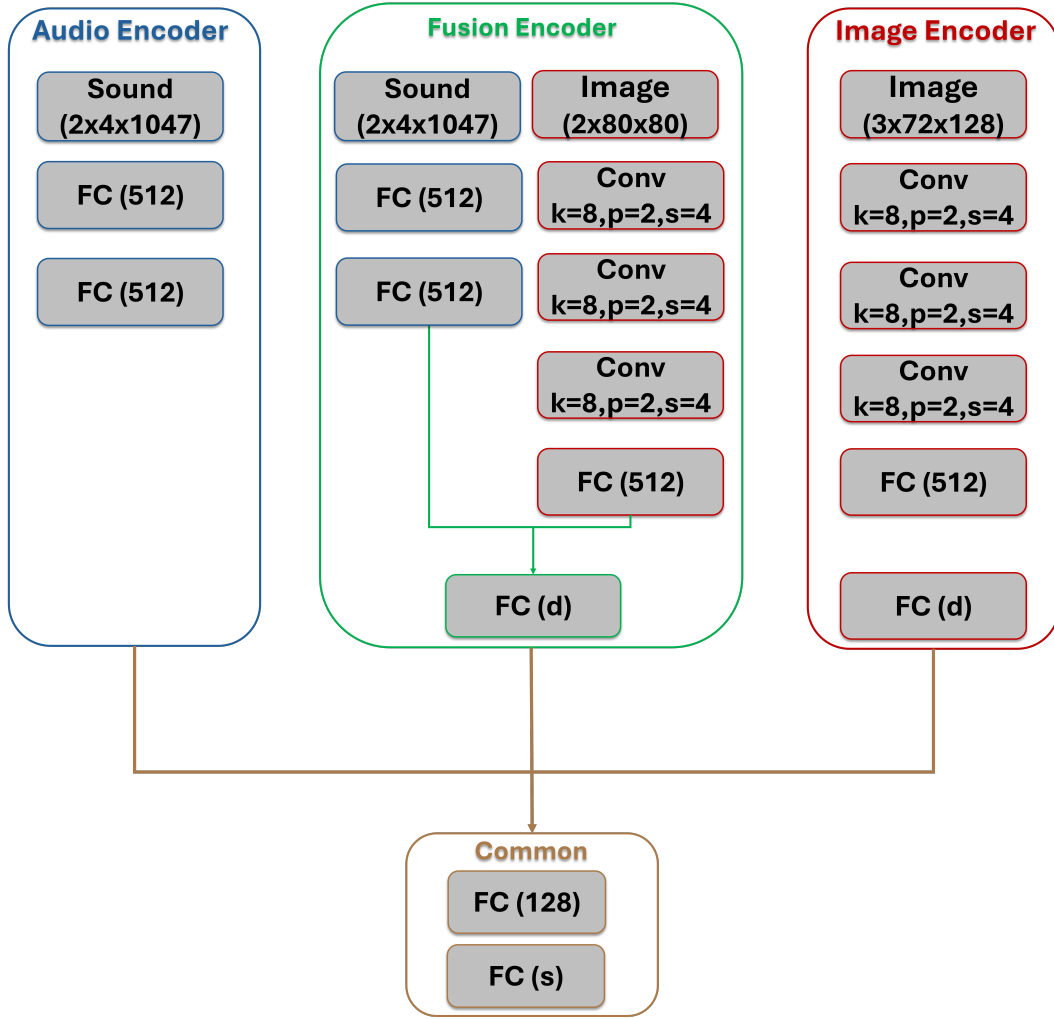


Figure F2: Multimodal encoder architecture used in HYPERHOT

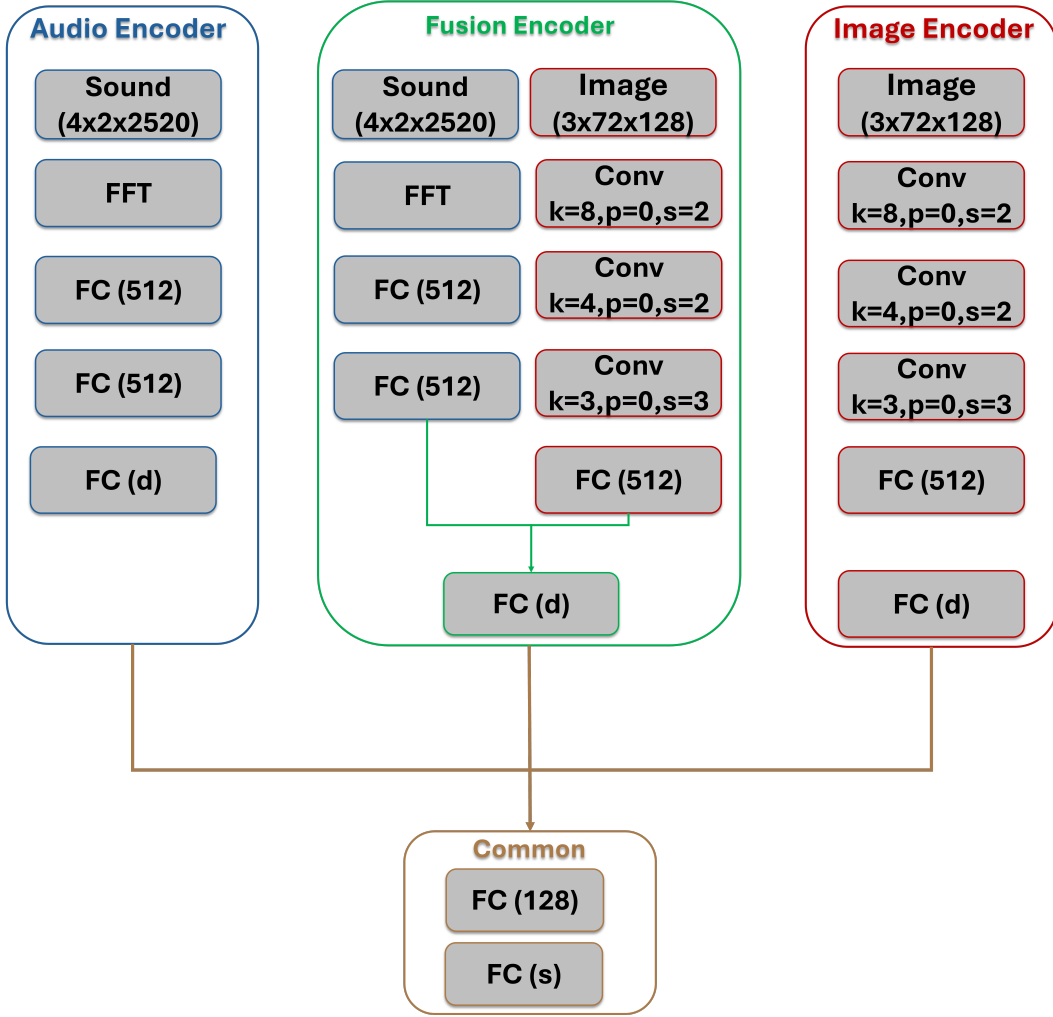


Figure F3: Multimodal encoder architecture used in VizDoom

A.2 Hyperparameter Settings

Table T1 shows hyperparameter settings used to train the multimodal encoder across the three games. Tables T2, T3, T4 show the hyperparameter settings to train RL in Pendulum, Hyperhot, and VizDoom, respectively.

Table T1: Multimodal encoder training hyperparameters in PENDULUM, HYPERHOT, and VizDoom.

Parameter	PENDULUM	HYPERHOT	VizDoom
Intermediate size d	64	64	64
Latent size s	10	10	10
Model training epochs	500	250	250
Learning rate	1e-3 (Decay)	1e-3 (Decay)	1e-3 (Decay)
Batch size B	128	128	128
Temperature τ	0.3	0.3	0.3

Table T2: Hyperparameters of Deep Deterministic Policy Gradient (DDPG) algorithm used to train RL agent in PENDULUM.

Hyperparameter	Value
Batch size d	128
$\eta_{actor}, \eta_{critic}$	1e-4, 1e-3
γ	0.99
Max episode length	300 frames
Replay buffer	25000
Max frames	150000
τ	1e-3

Table T3: Hyperparameters of Deep Q-Network (DQN) algorithm used to train the RL agent in HYPERHOT.

Hyperparameter	Value
Batch size d	128
η	1e-5
γ	0.99
Max episode length	450 frames
Replay buffer	350000
Max frames	350000

Table T4: Hyperparameters of Proximal Policy Optimization (PPO) algorithm used to train the RL agent in VizDoom.

Hyperparameter	Value
Batch size d	1024
γ	0.99
Max episode length	300 frames
Rollout	32
Frameskip	4
Training steps	5e8